

Achyut Tiwari

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RESEARCH PROFILE

Experimental condensed-matter physicist studying how electronic order and charge dynamics emerge in correlated and layered quantum materials. During my PhD, I used polarization-resolved infrared spectroscopy and Mueller-matrix ellipsometry to investigate charge dynamics, phase coexistence and interlayer-driven electronic states in correlated materials, particularly 1T-TaS₂ and organic charge-transfer salts. My experience includes cryogenic optical instrumentation, plasmonic and hyperbolic materials, thin-film preparation, and collaborative ultrafast pump-probe studies of light-induced phase transitions. I am particularly interested in extending this background toward nonequilibrium control of symmetry-broken and chiral states using ultrafast infrared and terahertz excitation.

PUBLICATIONS

- *Unconventional anisotropic charge dynamics in bulk 1T-TaS₂ induced by interlayer dimerization*
A. Tiwari, M. Wenzel, R. M. Roy, C. Prange, B. Gompf, and M. Dressel; **Editor's Suggestion**, *Phys. Rev. Lett.* (2026), (In press). [arXiv:2512.23883](#)
- *Interlayer coupling driven phase evolution in hyperbolic 1T-TaS₂ revealed by ellipsometry*
A. Tiwari, B. Gompf, and M. Dressel; **Editor's Pick**, *Appl. Phys. Lett.* **128**, 063104 (2026)
- *Nature of electronic phase transitions in substituted α -(BEDT-TTF)₂I₃ studied by generalized ellipsometry*
A. Tiwari, B. Gompf, S. Priya, D. Schweitzer, R. Kato, K. Hiraki, and M. Dressel; **Editor's Suggestion**, *Phys. Rev. B* **111**, 195142 (2025)
- *Temperature-dependent generalized ellipsometry of the metal-insulator phase transition in low-symmetry charge-transfer salts*
A. Tiwari, B. Gompf, and M. Dressel; **Featured Article**, *Appl. Phys. Lett.* **125**, 133101 (2024)
- *Ag_{100-x}Au_x Alloy Films: An Approach to Fixing Silver for Durable, Low-Loss Plasmonics*
B. Ewald, L. Siebigs, C. Zhang, J. Graf, **A. Tiwari**, M. Rödel, S. Hammer, V. Stepanenko, F. Würthner, B. Gompf, B. Hecht, J. Pflaum; *ACS Photonics* **12**, 6161 (2025)
- *Fermi-liquid behavior of non-altermagnetic RuO₂*
M. Wenzel, E. Uykur, S. Rößler, M. Schmidt, O. Janson, **A. Tiwari**, M. Dressel, A. A. Tsirlin; *Phys. Rev. B (Letter)* **111**, L041115 (2025)
- *Mixed Mott-Hubbard and charge transfer nature of 4H-SrMnO₃ thin film on Si (100)*
A. K. Mandal, A. Jana, S. Chowdhury, **A. Tiwari**, R.J. Choudhary, D.M. Phase; *J. Phys.: Condens. Matter* **33** 235501 (2021)
- *Natural hyperbolic dispersion in anisotropic bulk and nanoscaled copper chalcogenides*
U. Parekh, **A. Tiwari**, N. Didukh,..., B. Gompf, C. Klinke, R. Lesyuk; (Under review in Adv. Mat.)
- *Real-time tracking of instantaneous light induced phase transitions*
P. Kumar, S. Mandal, **A. Tiwari**,..., M. Dressel, E. Goulielmakis; (in preparation)
- *Effect of electron doping (Cerium-Ce) in epitaxial strain SrMnO₃ thin films*
A. K. Mandal, **A. Tiwari** et al., (in preparation)
- *Temperature, pressure and light tuning of the CDW gap and charge transfer in 6R-TaS₂*
R. M. Roy, M. Povolotsky, M. Wenzel, **A. Tiwari** et al., (in preparation)

RESEARCH EXPERIENCE

DOCTORAL RESEARCHER

2022-2026

1. PHYSIKALISCHES INSTITUT, UNIVERSITÄT STUTTGART, GERMANY

- Revealed a stacking-dependent, interlayer-dimerization-driven gap in a correlated van der Waals material 1T-TaS₂ by combining infrared spectroscopy with DFT, disentangling interlayer effects from in-plane CDW reconstruction. [Phys. Rev. Lett.](#), (2026)
- Extracted the full temperature-dependent dielectric tensor of triclinic crystals from azimuth- and angle-resolved Mueller-matrix ellipsometry, using an eigenpolarizability-based model to track its evolution across metal-insulator and charge-order transitions. ([Appl. Phys. Lett.](#) **125**, 133101)
- Developed quantitative optical models to distinguish intrinsic electronic reconstruction from spatial phase coexistence in correlated materials including 1T-TaS₂, α -(BEDT-TTF)₂I₃, V₂O₃. ([Phys. Rev. B](#) **111**, 195142; [Appl. Phys. Lett.](#) **128**, 063104)
- Integrated a continuous-flow He cryostat into the ellipsometry setup and developed stable alignment and calibration routines for cryogenic, polarization-resolved measurements.
- Developing magneto-optical Mueller-matrix ellipsometry to detect time-reversal-symmetry-breaking signatures in magnetic crystals (altermagnets), resolving weak polarization signatures of magnetic and electronic symmetry breaking

JUNIOR RESEARCH FELLOW

2021-2022

INDIAN INSTITUTE OF TECHNOLOGY (IIT) DELHI, INDIA

- Project: Optical characterization of graphene-based THz modulator on Si/HfO₂ heterostructures.

MASTER'S THESIS

2021

INDIAN INSTITUTE OF TECHNOLOGY (IIT) DHANBAD, INDIA

- Thesis: Surface-plasmon-resonance ellipsometry for refractive-index sensing; Au/Ag-based multilayer thin-film modeling for label-free biomolecular sensing applications.

RESEARCH PROJECTS

2018 & 2019

UGC-DAE CONSORTIUM FOR SCIENTIFIC RESEARCH, INDORE, INDIA

- SrMnO₃ thin films: pulsed-laser deposition on Si; HR-XRD, and resistivity measurements for structural and transport properties. ([J. Phys.: Condens. Matter](#) **33** 235501)
- BaTiO₃: solid-state synthesis; structural and electronic characterization using XRD and synchrotron XPS (Indus-1, BL-02).

EDUCATION

UNIVERSITÄT STUTTGART | PHD IN PHYSICS

2022-2026

- Advisor: [Prof. Dr. Martin Dressel](#)

IIT-DHANBAD | MSc IN PHYSICS

2019-2021

DEVI AHILYA UNIVERSITY, INDIA | BSc IN PHYSICS

2016-2019

EXPERIMENTAL TECHNIQUES & TECHNICAL SKILLS

- **Optical spectroscopy** : Polarization-resolved spectroscopy; FTIR; Mueller-matrix ellipsometry; Effective medium theory
- **Materials** : Correlated layered materials; organic conductors; transition-metal oxides; van der Waals materials; thin films
- **Low-temperature optical measurements** : Continuous-flow and closed-cycle cryostat
- **Electrical & structural characterization** : Transport measurements (PPMS); X-ray diffraction
- **Thin-film growth & sample preparation** : Pulsed laser deposition (PLD); cleanroom-based sample processing and electrical contact fabrication
- **Synchrotron user experience** : X-ray photoelectron spectroscopy (XPS) at synchrotron beamlines
- **Scientific tools** : MATLAB; Python; RefFIT; OriginPro; LaTeX; Inkscape; Blender

TEACHING EXPERIENCE

- Teaching Assistant (Advanced Physics Lab), MSc Physics, Universität Stuttgart. 2023–2025
Supervised experiments, data acquisition, and interpretation; provided feedback on lab reports
- Teaching Assistant (Engineering Physics Lab), BTech, IIT Delhi 2021–2022
Assisted in coordinating lab sessions, including setup checks and basic troubleshooting.

CONFERENCES AND WORKSHOPS

- 13th Workshop on Spectroscopic Ellipsometry (Genova, Italy) 2026
Presented a talk on [Phys. Rev. Lett. \(2026\)](#)
- DPG Spring Meeting of the Condensed Matter Section (Regensburg) 2025
Presented a poster on [Appl. Phys. Lett. 128, 063104 \(2026\)](#)
- DPG Spring Meeting of the Condensed Matter Section (Berlin) 2024
Presented a talk on [Phys. Rev. B 111, 195142](#)
- 12th Workshop on Spectroscopic Ellipsometry (Prague, Czech Republic) 2023
Presented a poster on [Appl. Phys. Lett. 125, 133101](#)

HONORS & AWARDS

- [CSIR-NET \(JRF\)](#) 2022
Qualified national exam for Junior Research Fellowship / Assistant Professor eligibility
- Junior Research Fellowship (IIT-Delhi)
Provided by IIT-Delhi as an institute fellow.
- [INSPIRE Scholarship](#) (Department of Science & Technology, Govt. of India) 2016-2021
Merit-based scholarship with monthly stipend and annual contingency grant (BSc–MSc).
- [National Graduate Physics Examination \(NGPE 2019\)](#): State Topper (Madhya Pradesh) 2019
State-level top rank.

REFERENCES

- [Prof. Dr. Martin Dressel](#) dressel@pi1.physik.uni-stuttgart.de
Professor,
1. Physikalisches Institut, Universität Stuttgart, Germany
- [Prof. Dr. Mathias Schubert](#) mschubert4@unl.edu
J.A.Woollam Distinguished Professor,
University of Nebraska-Lincoln (USA) & Lund University (Sweden)
- [Dr R. J. Choudhary](#) ram@csr.res.in
Scientist-H,
UGC – DAE Consortium for Scientific Research, Indore, India